MathDB 367323: all multiples of three, with the exact signs

Principia Math In Progress

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Task 1: Complete intended determinant formulas

Write $M(z) = 1 + zM(z) + z^2M(z)^2$ and

$$H_N^{(r)} = \det\big([z^{i+j}]M(z)^r\big)_{0\le i,j<N}, \qquad H_0^{(r)} = 1.$$

Wang and Zhang's Conjecture 3 (arXiv:2502.21050, p. 2) has a sign error: $H_9^{(9)} = 256$, whereas its printed sign gives -256. We prove both intended all-parameter formulas, including their precise signs. The result is substantially more than this isolated correction.

Theorem 1. *Let $r \ge 3$ be divisible by three and put $\epsilon_r = (-1)^{\binom{r}{2}}$. For every integer $n \ge 0$,*

$$H_{rn}^{(r)} = H_{rn+1}^{(r)} = \epsilon_r^n (n+1)^{r-1}, \tag{1}$$

$$H_{rn-1}^{(r)} + H_{rn+2}^{(r)} = -\frac{r(r-3)}{2}\epsilon_r^n (n+1)^{r-1}. \tag{2}$$

For $n = 0$ the second identity uses the harmless convention $H_{-1}^{(r)} = 0$; without it, take $n \ge 1$ there.

Thus the absolute value required of the source's α is correct, with $\alpha = -\epsilon_r^n r(r-3)/18$. The sign in (1) is $(-1)^n$ for $r \equiv 3, 6 \pmod{12}$ and 1 for $r \equiv 0, 9 \pmod{12}$.

We prove a stronger polynomial identity. For a real parameter a, let $M_a(z) = 1 + azM_a(z) + z^2M_a(z)^2$, and denote the corresponding determinants by $H_N^{(r)}(a)$. Define

$$P_j(t) = U_j\left(\frac{t-a}{2}\right), \qquad A(t) = P_{r-1}(t), \qquad B_k(t) = U_k\left(T_r\left(\frac{t-a}{2}\right)\right),$$

where $T_j(\cos\theta) = \cos(j\theta)$ and $U_j(\cos\theta) = \sin((j+1)\theta)/\sin\theta$. The polynomials P_j and B_k are monic, of degrees j and rk.

Theorem 2 (Weighted block boundary identities)**.** *For every $r \ge 2$, $k \ge 0$, and real or complex a, with $N = rk$,*

$$H_N^{(r)}(a) = H_{N+1}^{(r)}(a) = \epsilon_r^k B_k(0)^{r-1}, \tag{3}$$

$$H_{N-1}^{(r)}(a) + H_{N+2}^{(r)}(a) = \left(r - \binom{r}{2}a^2\right) H_N^{(r)}(a). \tag{4}$$

Again $H_{-1}^{(r)}(a) = 0$.

The convolution functional and its determinant

For real a, let μ_a be the translated semicircle measure

$$d\mu_a(t) = \frac{\sqrt{4-(t-a)^2}}{2\pi}\,dt \quad (a-2 \le t \le a+2).$$

Its moments are $[z^j]M_a(z)$, and its monic orthogonal polynomials are P_j, with $\int P_i P_j \, d\mu_a = \delta_{ij}$. Both statements follow by $t = a + 2\cos\theta$: the centered moments are the Catalan numbers and the orthogonality integral is the sine integral. Set $Q(t) = t^{r-1}A(t)$ and $L(f) = \int f(t)Q(t)\, d\mu_a(t)$. Then

$$[z^j]M_a(z)^r = L(t^j). \tag{5}$$

Indeed $G(w) = w^{-1}M_a(w^{-1})$ satisfies $G^2 = (w-a)G - 1$, so induction gives $G^r = P_{r-1}(w)G - P_{r-2}(w)$. Multiplication by w^{r-1} and extraction of w^{-j-1} gives (5). The functional L need not be positive.

We use the following classical Christoffel identity in an algebraic form. Its evaluation form is Theorem 1 of Krattenthaler, arXiv:2101.04225, with repeated roots covered by Proposition 5. A proof is included to make the argument self-contained.

Lemma 1. *If Q is monic of even degree d, the determinant of the moments of $Q\, d\mu_a$ of order m is the determinant of the $d \times d$ matrix whose jth column is the coefficient vector of $P_{m+j} \bmod Q$, for $0 \le j < d$.*

Proof. First suppose Q has distinct roots $\alpha_1, \dots, \alpha_d$. Expansion of determinants gives the moment determinant as

$$\frac{1}{m!} \int \cdots \int \Delta(x)^2 \prod_{i=1}^{m} Q(x_i) \prod_{i=1}^{m} d\mu_a(x_i).$$

The mixed Vandermonde satisfies $\Delta(x,\alpha) = \Delta(x)\Delta(\alpha)\prod_{i,j}(\alpha_j - x_i)$. Replace monomials in both Vandermondes by the monic P_j. Orthogonality leaves only columns $0, \dots, m-1$ for the x rows, with integral $m!$. Hence the determinant is

$$(-1)^{md} \frac{\det(P_{m+j}(\alpha_i))_{1\le i\le d,\, 0\le j<d}}{\det(\alpha_i^j)_{1\le i\le d,\, 0\le j<d}}.$$

The sign is one. Evaluation of remainders factors the numerator matrix as the denominator matrix times the coefficient matrix. The identity extends to repeated roots because both sides are polynomials in the coefficients of the monic Q; polynomial division introduces no denominators. It also includes $m = 0$. □

Exact evaluation at a block boundary

Put $N = rk$ and $c = N + r - 1$. The sine formula gives

$$P_c = AB_k. \tag{6}$$

In the remainder determinant of Lemma 1, retain its first $r-1$ columns $P_N, \dots, P_{N+r-2}$. For each $1 \le j \le r-2$, add column P_{c-j} to column P_{c+j}. The last $r-1$ columns then become

$$AB_kS_0, \dots, AB_kS_{r-2}, \qquad S_0 = 1, \quad S_j(t) = 2T_j((t-a)/2) \ (j \ge 1).$$

Here $P_{c+j} + P_{c-j} = S_jP_c$, and each S_j is monic of degree j.

Use the following monic basis of $\mathbb{C}[t]/(t^{r-1}A)$:

$$P_0, \dots, P_{r-2}, \quad AS_0, \dots, AS_{r-2}.$$

Its change-of-basis determinant from the monomials is one. Reduction modulo A gives an upper block with the first $r-1$ columns and zero in the last columns. At the simple roots $\rho_j = a + 2\cos(j\pi/r)$ of A,

$$P_{N+h}(\rho_j) = (-1)^{kj}P_h(\rho_j) \qquad (0 \le h \le r-2).$$

Thus the upper block determinant is $\prod_{j=1}^{r-1}(-1)^{kj} = \epsilon_r^k$. The lower right block represents multiplication by B_k in $\mathbb{C}[t]/(t^{r-1})$. In its monomial basis this matrix is triangular, with diagonal $B_k(0)$, so its determinant is $B_k(0)^{r-1}$. This proves the formula for H_N in (3). No coprimality of A and t was assumed.

The adjacent determinant and the paired sum

Temporarily choose a real a for which $B_k(0) \neq 0$. This excludes only finitely many values. Equation (6) gives

$$L(t^j B_k) = \int t^{j+r-1} P_{N+r-1}(t)\, d\mu_a(t) = 0 \qquad (0 \le j < N).$$

Consequently $q_N = B_k$ is the unique monic polynomial of degree N orthogonal to lower degrees for L, since $H_N \neq 0$. The elementary identity $U_k(x)^2 = \sum_{h=0}^{k} U_{2h}(x)$ yields

$$AB_k^2 = \sum_{h=0}^{k} P_{(2h+1)r-1}. \tag{7}$$

For $j = 0, 1, 2$, every $h \ge 1$ term is orthogonal to t^{r-1+j}, because $r \ge 2$. Therefore

$$L(q_N^2) = 1, \qquad L(tq_N^2) = ra, \qquad L(t^2 q_N^2) = r + \binom{r+1}{2} a^2. \tag{8}$$

For completeness, the last two values are the coefficients of P_{r-1} in t^r and t^{r+1}. The recurrence $tP_j = P_{j+1} + aP_j + P_{j-1}$, with $P_{-1} = 0$, gives respectively ra and $\binom{r+1}{2}a^2 + r$: in the latter coefficient there are two level steps or one down step among the otherwise up steps.

The Gram determinant identity $L(q_N^2) = H_{N+1}/H_N$ now proves $H_{N+1} = H_N$. In particular q_{N+1} also exists. Suppose first that $N \ge 1$. Let $G_N = (L(t^{i+j}))_{0 \le i,j < N}$ and write

$$tq_N = q_{N+1} + ra\, q_N + v, \qquad \deg v < N.$$

For $j < N - 1$, $L(t^j v) = 0$, whereas $L(t^{N-1} v) = 1$. Thus the coefficient vector of v is $G_N^{-1} e_{N-1}$, and

$$L(v^2) = (G_N^{-1})_{N-1,N-1} = \frac{H_{N-1}}{H_N}.$$

The three summands in the displayed decomposition are mutually orthogonal. Taking squared norms and using (8) gives

$$r + \binom{r+1}{2} a^2 = \frac{H_{N+2}}{H_{N+1}} + r^2 a^2 + \frac{H_{N-1}}{H_N},$$

which is (4). This argument never divides by H_{N-1} or H_{N+2}, which are allowed to vanish. For $N = 0$, the same identity is directly $H_2 = L(t^2) - L(t)^2 = r - \binom{r}{2} a^2$. Finally, the asserted identities are polynomial identities in a, so they extend from the generic real values to every complex value, including $B_k(0) = 0$. This proves Theorem 2.

Specialization to the conjecture

Set $a = 1$, $k = n$, and $3 \mid r$. Since $T_r(-1/2) = \cos(2\pi r/3) = 1$, we have $B_n(0) = U_n(1) = n + 1$. Theorem 2 now gives (1)–(2). For the explicit sign discrepancy, $r = 9, n = 1$ gives $\epsilon_9 = 1$ and $H_9 = H_{10} = 2^8 = 256$.

Task 2: Sharp degrees in the remaining residue classes

Determine the degrees and leading coefficients of the polynomial blocks $H_{rn+s}^{(r)}$ for $2 \le s < r$, including systematic zeros.

Task 3: Separate the paired determinants

Find closed formulas for $H_{rn-1}^{(r)}$ and $H_{rn+2}^{(r)}$ individually, refining their simple sum.

Task 4: Other weights with exceptional boundaries

Classify the real parameters a for which $B_k(0)$ vanishes for infinitely many k, and determine the neighboring nonzero determinants.

Task 5: A path interpretation of the exact magnitudes

Seek a cancellation argument on families of weighted Motzkin paths that explains the factor $(n+1)^{r-1}$ and the sign ϵ_r^n.